

NAZMUL HUDA BADHON

Dhaka, Bangladesh, badhon15-4772@diu.edu.bd, +880 1903 906230

[LinkedIn](#), [Google Scholar](#), [Portfolio](#)



Graduate with peer-reviewed publications in deep learning and machine learning, with experience in convolutional neural networks. International academic exposure through a funded exchange study.

EDUCATION

Bachelor of Science	Computer Science and Engineering Daffodil International University CGPA: 3.88 / 4.00	Spring 2022–Fall 2025 Dhaka, Bangladesh
	Thesis: "Tabular data-based lightweight convolutional neural network for electricity energy demand prediction."	
	Computer Engineering (Exchange Semester) Dongseo University, GPA: 3.94 / 4.50	Fall 2023 Busan, South Korea

RESEARCH INTERESTS

Designing computationally efficient deep neural network architectures.
Developing robust and generalizable deep learning models for real-world applications.

RESEARCH PUBLICATIONS

Research Articles

1. **Nazmul Huda Badhon**, Imrus Salehin, Md. Tomal Ahmed Sajib, Md. Sakibul Hassan Rifat, S M Noman, Nazmun Nessa Moon, "[TLCNN: tabular data-based lightweight convolutional neural network for electricity energy demand prediction.](#)" Global Energy Interconnection (Q1), Elsevier, 14 Nov 2025. ISSN: 2590-0358.
2. **Nazmul Huda Badhon**, Sabit Ahmed Preanto, Abu Shahed Shah Md Nazmul Arefin, Kazi Jahid Hasan, and Md. Baharul Islam. "[Temporal convolutional network based indoor location recognition from ble beacon signals in nursing care facility.](#)" International Journal of Activity and Behavior Computing, 13 Apr 2026. ISSN: 2759-2871.
3. Md. Tomal Ahmed Sajib, **Nazmul Huda Badhon**, Imrus Salehin, Md. Sakibul Hassan Rifat, Faysal Ahmmed, Nazmun Nessa Moon, "[A comparative deep learning methodology for plant insect image classification: assessment of CNN architectures and augmentation techniques.](#)" MethodsX (Q2), 103680, Elsevier, 17 Oct 2025. ISSN: 2215-0161.

Systematic Review

4. Imrus Salehin, Md. Tomal Ahmed Sajib, **Nazmul Huda Badhon**, Md. Sakibul Hassan Rifat, Nazrul Amin, Nazmun Nessa Moon, "[Systematic Literature Review of LLM-Large Language Model in Medical: Digital Health, Technology and Applications.](#)" Engineering Reports (Q2), 7.9: e70365, Wiley, 12 Sep 2025. ISSN: 2577-8196.

Data Article

5. Imrus Salehin, **Nazmul Huda Badhon**, Md Tomal Ahmed Sajib, Mohd Saifuzzaman, and Nazmun Nessa Moon. "IQ-UltraRecon: Demodulated IQ ultrasound dataset of human hand and arm tissue for deep learning-based reconstruction." Data in Brief (Q3), 112001, Elsevier, 22 Aug 2025. ISSN: 2352-3409.
6. Imrus Salehin, **Nazmul Huda Badhon**, Md Tomal Ahmed Sajib, and Nazmun Nessa Moon. "DeepUS-ReconSeg: A multi-angle paired B-mode Ultrasound dataset for medical imaging reconstruction and segmentation." Data in Brief (Q3), 112083, Elsevier, 19 Sep 2025. ISSN: 2352-3409.
7. Imrus Salehin, Mahbubur Rahman Khan, Ummya Habiba, **Nazmul Huda Badhon**, Nazmun Nessa Moon. "BAU-Insectv2: An Agricultural Plant Insect Dataset for Deep Learning and Biomedical Image Analysis." Data in Brief (Q3), 110083, Elsevier, 23 Jan 2024. ISSN: 2352-3409.

Book Chapter

8. Imrus Salehin, **Nazmul Huda Badhon**, Md. Tomal Ahmed Sajib, Md. Sakibul Hassan Rifat, Nazmun Nessa Moon. Predicting Peak Energy Demand Using Machine Learning Techniques for Efficient Electricity Supply Management. Engineering Applications of AI for Demand Forecasting, Taylor & Francis. Accepted on 24 Apr 2025 (In press).

PROJECTS

Selected Research Projects

1. **Breast Cancer Classification** using Hybrid Segmentation Deep Learning
 - Built a two-stage U-Net and CNN-based hybrid model on the BUSI datasets.
 - Achieved strong performance on accuracy and ROC metrics.
2. **Lung and Colon Cancer Histopathology** Classification
 - Applied preprocessing on histopathology and evaluated transfer learning models.
 - Identified the best-performing model by accuracy and F1-score.
3. **Vision Transformer**-Based Medical Image Classification
 - Designed preprocessing workflows for medical images and classification with ViT.
 - Achieved 87% testing accuracy and improved transformer performance.

Additional Projects

4. **Lost and Found Web Application**
 - Developed a full-stack web app with HTML, CSS, and JS for reporting and searching lost and found items.
 - Implemented PHP-based backend, REST APIs, MySQL database, and secure user/admin authentication with image upload support.

SCHOLARSHIPS AND AWARDS

Global Korea Scholarship (GKS)—fully funded exchange scholarship **Fall 2023**
Govt. of the Republic of Korea

Special Scholarship for Consistent Academic Excellence **Spring 2022–Fall 2025**
Daffodil International University

Research Award for Outstanding Undergraduate Research Contribution **2024**
Division of Research, Daffodil International University

Technical skills

- **Machine Learning & Deep Learning:**
Convolutional Neural Networks, U-Net, Vision Transformers, Transfer Learning
- **Frameworks & Libraries:** PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV
- **Programming:** Python, Java, C
- **Data Analysis & Visualization:** NumPy, Pandas, Matplotlib, Seaborn
- **Tools & Platforms:** GitHub, Google Colab, LaTeX

ACADEMIC SERVICE & LEADERSHIP

Peer Reviewer Recognition **2025**
Journal: Digital Health Publisher: SAGE Publishing

Volunteer Internship **2024**
International Affairs, Daffodil International University

Team Leader **2023**
International Camping, Dongseo University, Busan, South Korea

REFERENCES

Nazmun Nessa Moon Associate Professor, Dept. of Computer Science & Engineering, Daffodil International University, Dhaka, Bangladesh Email: moon@daffodilvarsity.edu.bd	Dr. Fernaz Narin Nur Professor, Dept. of Computer Science & Engineering, Daffodil International University, Dhaka, Bangladesh Email: fernaz.cse@diu.edu.bd
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